

WebRTC Browser Calling — Phase 2

As-Built Documentation

browser-calling-5194.twil.io | May 2025

Overview

Phase 2 extends the Phase 1 browser calling foundation with three capabilities: full attribution of calls in HubSpot, a spoken whisper to the agent before the customer is bridged, and real-time call transcription appended to the HubSpot engagement. This document describes all components as deployed.

 Status	Fully deployed and operational as of May 2025. All components tested end-to-end.
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System Architecture

Component	Type	Location	Purpose
call-page.html	Static HTML	Twilio Asset /index.html	Customer-facing call page
generate-token.js	Twilio Function	/generate-token (Public)	Issues signed Access Tokens with shared secret validation
voice-handler.js	Twilio Function	/voice-handler (Public)	Routes call, starts transcription, whisper, status callback
whisper-handler.js	Twilio Function	/whisper-handler (Public)	Returns TwiML whisper to agent before bridging
transcript-handler.js	Twilio Function	/transcript-handler (Public)	Accumulates transcript chunks in Sync, stores on completion
CallLogger.gs	Google Apps Script	GAS Web App (doPost/doGet)	HubSpot logging, transcript retrieval from Sync
TestSuite.gs	Google Apps Script	GAS Editor (manual run only)	Test functions for all integration components
Twilio Sync	Twilio Service	Console → Sync → Services	Temporary transcript storage between transcript-handler and GAS

Complete Call Flow

1. Customer opens the call page URL in their browser — e.g. <https://browser-calling-5194.twil.io/index.html?ref=BK105067&name=Mark+Lonergan>

2. Page sends GET to generate-token with bookingRef, customerName, and TOKEN_SECRET. Function validates the secret and issues a signed JWT Access Token.
3. Browser initialises Twilio.Device with the token. Status pill shows 'Line open'.
4. Customer taps Call Now — Twilio.Device.connect() sends the call to Twilio's cloud with bookingRef and customerName as custom params.
5. Twilio calls voice-handler. Function returns raw XML TwiML containing:
<Start><Transcription> pointing at transcript-handler, <Dial> pointing at Aircall with answerOnBridge=true, whisper URL on <Number>, and action URL (GAS) on <Dial>.
6. Twilio starts real-time transcription on both audio tracks (customer + agent), streaming utterances to transcript-handler as they occur.
7. Aircall rings. Agent answers in Aircall. Twilio calls the whisper URL on the agent's leg before bridging.
8. whisper-handler returns TwiML <Say> spoken privately to the agent: 'Wilderness browser call. Customer: Mark Lonergan. Booking reference: BK105067. Connecting now.'
9. Customer and agent are bridged — call proceeds normally.
10. Call ends. Twilio fires two simultaneous events: (a) transcript-handler receives transcription-stopped and assembles the full transcript, storing it in Twilio Sync under transcript_{CallSid}. (b) GAS doPost receives the status callback.
11. GAS doPost logs the call to HubSpot as a Call engagement (with Call Type = Inbound Phone Call, Call Outcome = Connected, contact and deal associations). Writes HubSpot Call ID to the Call Log sheet. Schedules a GAS trigger to fire in 45 seconds.
12. 45 seconds later, the trigger fires runPendingTranscriptLookup(). This reads the transcript from Twilio Sync, appends it to the HubSpot Call engagement body, and cleans up the Sync document.

i Timing note

The 45-second trigger delay ensures transcript-handler has finished writing to Sync before GAS reads it. If Sync is not ready, the function retries up to 5 times at 10-second intervals (up to 95 seconds total).

Environment Variables — Full Reference

Twilio Function Service: browser-calling

Set in Twilio Console → Functions → browser-calling → Environment Variables. All changes require Deploy All.

Variable	Value / Format	Used by	Phase
ACCOUNT_SID	AC... (auto-available)	generate-token.js	1
API_KEY_SID	SK...	generate-token.js	1
API_KEY_SECRET	Secret from API Key creation	generate-token.js	1
TWIML_APP_SID	AP...	generate-token.js	1

TOKEN_SECRET	32-char random string	generate-token.js	2
AIRCALL_NUMBER	+64XXXXXXXXXX (E.164)	voice-handler.js	1
TWILIO_NUMBER	+64XXXXXXXXXX (E.164)	voice-handler.js	1
GAS_WEBHOOK_URL	https://script.google.com/macros/s/.../exec	voice-handler.js	2
WHISPER_HANDLER_URL	https://browser-calling-5194.twil.io/whisper-handler	voice-handler.js	2
TRANSCRIPT_HANDLER_URL	https://browser-calling-5194.twil.io/transcript-handler	voice-handler.js	2
SYNC_SERVICE_SID	IS...	transcript-handler.js	2

Google Apps Script: Script Properties

Set in GAS Editor → Project Settings → Script Properties.

Property	Value	Used by
HUBSPOT_OWNER_ID	HubSpot user ID (optional)	CallLogger.gs — assigns call engagements to a default owner
ALERT_EMAIL	Email address for daily monitoring	CallLogger.gs — monitorCallLog() alert recipient
TWILIO_ACCOUNT_SID	AC... (same as Twilio Console)	CallLogger.gs — authenticates Twilio Sync API calls
TWILIO_AUTH_TOKEN	Auth token from Twilio Console	CallLogger.gs — authenticates Twilio Sync API calls
SYNC_SERVICE_SID	IS... (same as Twilio env var)	CallLogger.gs — reads transcript from Sync documents

HubSpot Call Engagement — Properties Set

HubSpot Property	Value logged	Notes
hs_call_title	Browser Call — {customerName}	
hs_call_body	Source, status, duration, booking ref, Twilio SID, Deal ID, transcript	Transcript appended ~45s after call ends
hs_call_status	COMPLETED / NO_ANSWER / BUSY / FAILED	Maps from Twilio DialCallStatus
hs_call_direction	INBOUND	
hs_call_disposition	Connected / No answer / Busy / Wrong number	GUID from /calling/v1/dispositions

hs_activity_type	Inbound Phone Call	Sets Call Type dropdown
hs_call_duration	Duration in milliseconds	
hs_call_from_number	WebRTC Browser	
hs_timestamp	Call start time	
Associations	Contact (194) + Deal (206)	Looked up via Deal name = booking ref → Deal → Contact association

Feature Flags

Defined at the top of CallLogger.gs. Toggle without redeploying Twilio — redeploy GAS as a new version only.

Flag	Default	Effect when false
WHISPER_ENABLED	true	GAS returns empty customerName to whisper-handler — agent hears no name
TRANSCRIPT_ENABLED	true	Transcript not appended to HubSpot; runPendingTranscriptLookup skipped
HUBSPOT_LOGGING	true	No HubSpot Call engagement created
SHEET_LOGGING	true	No row written to Call Log sheet

Token Security

generate-token.js validates a shared secret before issuing any Access Token. Requests without a valid secret receive HTTP 403.

- TOKEN_SECRET env var set in Twilio Functions environment variables
- Same value hardcoded in call-page.html as TOKEN_SECRET JavaScript constant
- To rotate: generate a new secret at generate-secret.vercel.app/32, update both locations, Deploy All in Twilio, re-upload call-page.html



Note

The secret is visible in the page source. The blast radius of abuse is limited — a discovered secret can only be used to generate tokens that place calls to your Aircall number at your Twilio cost. Rotate immediately if abuse is suspected.

Monitoring & Alerting

A daily trigger runs `monitorCallLog()` at 8am NZT and emails `ALERT_EMAIL` if any issues are detected in the past 24 hours.

Issue detected	Likely cause
HubSpot Status = Error	HubSpot API failure — check token scopes or API availability
Blank HubSpot Status	GAS doPost did not complete — check deployment version
No HubSpot Call ID	Call engagement was not created — see error column
No contact match	Deal not found for booking ref — check Rata → HubSpot sync
Missing booking ref	Call link was opened without <code>?ref=</code> parameter
Transcript not appended	Sync doc expired or trigger failed — check GAS trigger list

To install the trigger: run `setupMonitoringTrigger()` once from the GAS editor. To test alerts: run `testMonitoringEmail()`.

Call Log Sheet — Column Reference

Column	Header	Content
A	Timestamp	ISO timestamp when Twilio status callback was received
B	Call SID	Twilio parent CallSid — used as lookup key for transcript
C	Booking Ref	From <code>?ref=</code> URL parameter
D	Customer Name	Resolved from HubSpot if not in URL; from <code>?name=</code> otherwise
E	Duration (s)	DialCallDuration from Twilio callback
F	Dial Status	completed / no-answer / busy / failed
G	HubSpot Status	Logged ✓ / Logged ✓ + Transcript ✓ / Error / Logged (no match)
H	HubSpot Contact ID	ID of matched contact
I	HubSpot Call ID	ID of created Call engagement — used for transcript append
J	Caller	Twilio client identity string including booking ref
K	Error	Error message if any step failed

Known Constraints & Limitations

Constraint	Detail	Workaround
Caller ID in Aircall	All browser calls arrive showing the Twilio number, not the customer's number	Whisper announces customer name before bridging
Whisper timing	Agent hears the whisper while their phone is still ringing — first few words may be cut off if Aircall answers slowly	Whisper includes a 3-second pause and repeats the message twice
Transcript language	Real-time transcription uses en-US — NZ English accent generally transcribed well but occasional regional terms may be misrecognised	Review transcript in HubSpot after each call
Transcript delay	Transcript appears in HubSpot ~45–95 seconds after call ends, not immediately	By design — ensures Twilio Sync write completes before read
Token secret in page source	TOKEN_SECRET is visible in browser DevTools	Low risk at current volumes; rotate if abused
Aircall double logging	Aircall's HubSpot integration may also create a Call engagement for the same call	Check contact timeline — GAS engagement includes transcript and booking ref; Aircall's does not

Google Apps Script — Function Reference

CallLogger.gs — Production functions

Function	Trigger	Purpose
doGet(e)	HTTP GET from Twilio/whisper-handler	Returns customer name JSON for whisper lookup (?type=whisper-name-only)
doPost(e)	HTTP POST from Twilio status callback	Logs call to HubSpot, schedules transcript trigger
logCallToHubSpot_(params)	Called by doPost	Creates HubSpot Call engagement with all properties
findContactByDeal_(ref)	Called by logCallToHubSpot_	Searches deals by name → traverses association → returns contact ID
getCustomerNameFromDeal_(ref)	Called by doGet whisper handler	Resolves customer full name from HubSpot contact via deal
appendTranscriptFromSync_(sid)	Called by runPendingTranscriptLookup	Reads transcript from Twilio Sync, appends to HubSpot engagement

<code>scheduleTranscriptLookup_(s id)</code>	Called by doPost	Creates a 45-second one-off GAS trigger
<code>runPendingTranscriptLookup()</code>	GAS time-based trigger (45s delay)	Processes all pending transcript lookups from Script Properties
<code>monitorCallLog()</code>	GAS daily trigger (8am NZT)	Scans Call Log sheet, emails ALERT_EMAIL if issues found
<code>setupMonitoringTrigger()</code>	Manual — run once	Installs the daily monitorCallLog trigger

TestSuite.gs — Test functions (manual run only)

Function	Creates HubSpot data?	Purpose
<code>testRunAll()</code>	No	Runs all read-only tests in sequence
<code>testFeatureFlags()</code>	No	Prints current flag states
<code>testHubSpotConnectivity()</code>	No	Confirms token valid and API reachable
<code>testDealLookup()</code>	No	Tests deal → contact lookup chain
<code>testDealLookupMiss()</code>	No	Confirms graceful handling of bad booking ref
<code>testWhisperNameResolution()</code>	No	Tests HubSpot name resolution for whisper
<code>testTokenSecurity()</code>	No	Confirms 403 on bad/missing token secret
<code>testMonitorDryRun()</code>	No	Scans last 24h without sending email
<code>testCallLogging()</code>	Yes	Logs a simulated completed call to HubSpot
<code>testNoAnswerCallLogging()</code>	Yes	Logs a simulated no-answer call
<code>testTranscriptAppend()</code>	Yes	Appends fake transcript to last real call
<code>testFullCallLifecycle()</code>	Yes	Runs all steps end-to-end with real HubSpot writes
<code>testMonitoringEmail()</code>	No	Sends test alert email to ALERT_EMAIL
<code>testMonitoringTrigger()</code>	No	Confirms daily trigger is installed